Sarah M. Coleman, PhD

Open to relocation nationwide · Available to start within 30 days, at own expense
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Summary

Bioprocess development engineer specializing in **upstream process development** and **fed-batch fermentation**. Designed and implemented a first-in-company **fed-batch bioreactor** process end to end, covering feed strategy, control parameters, **SOPs**, **batch records**, and KPIs, and led **tech transfer** to manufacturing. Background spans strain and fermentation process development, multi-omics analysis, and statistical modeling, with upstream PD experience at **Bristol Myers Squibb** (biologics, mammalian **cell culture**) and **Takeda** (vaccines). PhD in Chemical Engineering and MS in Statistics: **DoE**, **scale-up/scale-down**, and statistical process modeling in Python and R.

Work Experience

BioReNuva, Austin, TX
Bioprocess Development Engineer (Contract/Fixed-term) | Jun 2026 – Present

- Engaged as subject-matter expert to establish fed-batch fermentation capability for glycolipid production using *Pseudomonas* strains, building on prior collaboration with the Alper Lab
- Designed and implemented the company's first fed-batch bioreactor process from scratch, including feed strategy, control parameters, and run protocols
- Authored the standard operating procedure (SOP) for bioreactor runs, establishing the reference process documentation used for ongoing and future fermentation campaigns
- Defined key performance indicators (KPIs) and identified process bottlenecks for glycolipid synthesis, informing the roadmap for scale-up and continuous improvement
- Led technology transfer activities, translating process knowledge into documentation and protocols to hand off from development to manufacturing teams

The University of Texas at Austin, Austin, TX
Postdoctoral Research Fellow, Statistics and Data Sciences | Oct 2024 – Sep 2026
Research Advisor: Dr. Roger D. Peng

- Developed epidemiological and statistical models of large environmental-health datasets, and advised medical faculty on statistical methodology and experimental design

Lecturer, Statistics and Data Sciences | Jan 2026 – Apr 2026

- Instructor of record for 100-student upper-division data science course, rated 4.32/5 overall by the students, a higher rating than the department, college, and university averages; managed 3 TAs

Graduate Student Fellow, Chemical Engineering | Aug 2019 – Aug 2024
Research Advisor: Dr. Hal S. Alper

- Integrated CFD modeling with fermentation experiments to characterize and optimize bioreactor performance in high oil density 3 L bioreactor systems
- Evolved *Yarrowia lipolytica* to convert industrial biomass waste into the high-value polyketides triacetic acid lactone and simultaneously reduce COD (Chemical Oxygen Demand)
- Evolved *Y. lipolytica* for growth on high-density waste cooking oil and engineered it to produce long-chain fatty alcohols, characterizing the evolved strain by genomics, transcriptomics, and metabolomics
- Developed a genome editing system using Golden Gate assembly for targeted engineering of *Y. lipolytica*, and used this system to screen a native promoter library
- Proposed and validated a statistical model to quantitatively evaluate the performance of biological Boolean logic gates for the first time

MathWorks, Remote (Natick, MA)
MATLAB Student Ambassador | Mar 2020 – Jul 2021

- Delivered technical workshops translating research and engineering workflows into MATLAB-based computational pipelines for academic users
- Demonstrated applied modeling and data analysis workflows, accelerating adoption of reproducible computational methods in research environments

Massachusetts Institute of Technology, Cambridge, MA
Advanced Undergraduate Researcher | Sep 2018 – Aug 2019

- Developed a method of nanophosphor size sorting to improve the capacitance of nanoparticles infiltrated into a light emitting plant
- Tested the influence of nanoparticle infiltration on chlorophyll concentration and photosynthetic ability

Undergraduate Researcher | Sep 2016 – Aug 2018

- Developed anaerobic assays to characterize the carbon assimilation of *Moorella thermoacetica* enzymes in the Wood-Ljungdahl pathway

Bristol Myers Squibb, Devens, MA
Intern, Upstream Process Development (Biologics) | Jun 2018 – Aug 2018
- Designed and executed metabolite-controlled fed-batch fermentation experiments to systematically evaluate monoclonal antibody production performance in mammalian cell lines
- Analyzed upstream bioprocess data to identify conditions that improved yield stability and process robustness in biologics production
- Supported experimental iteration cycles for process optimization in a regulated biomanufacturing environment

Takeda Pharmaceuticals, Cambridge, MA
Intern, Process Development (Vaccines) | May 2017 – Aug 2017
- Optimized mammalian cell culture conditions to improve vaccine production performance within a next-generation manufacturing system
- Contributed to experimental design and process optimization workflows focused on improving scalability and production efficiency of biologics

Girls' Angle Math Club, Cambridge, MA
Math Mentor | Sep 2016 – Apr 2018
- Mentored students weekly in advanced mathematical reasoning with emphasis on structured problem-solving and analytical thought processes

Education

The University of Texas at Austin, Austin, TX
Ph.D. in Chemical Engineering, Aug 2024
The University of Texas at Austin, Austin, TX
M.S. in Statistics, May 2024
Massachusetts Institute of Technology, Cambridge, MA
S.B. in Chemical-Biological Engineering, Jun 2019

Technical Skills

- **Bioprocessing & Fermentation**: Batch and fed-batch fermentation, bioreactor operation and control (BioFlo/CelliGen, DeltaV), dissolved oxygen and pH cascade control, aeration and agitation strategy, scale-up/scale-down, multiphase and high oil density fermentation, k_La and mass transfer analysis, media and feed development, 96-well plate screening, assay development, high-throughput screening workflows, SOPs, batch records, tech transfer
- **Metabolic & Strain Engineering**: Metabolic and pathway engineering, genome editing, Golden Gate assembly, homologous recombination, strain engineering, promoter library screening, directed/adaptive laboratory evolution
- **Cell Culture & Molecular Biology**: Yeast, bacterial and mammalian cell culture, aseptic technique, cGLP practices, qPCR, flow cytometry, protein expression & purification, Western blotting
- **Computational Biology & Data Science**: Python (pandas, NumPy, scikit-learn), R (tidyverse), SQL, MATLAB, JMP, statistical modeling, machine learning, experimental design (DoE), computational fluid dynamics (Ansys CFX), reproducible data analysis, pathway & enrichment analysis
- **Bioinformatics & Omics Workflows**: RNA-seq and Tag-seq workflows (cutadapt, STAR, HTSeq, DESeq2), genomic variant analysis (bowtie2, samtools/bcftools), genome annotation (AUGUSTUS) and orthology (BLAST), functional enrichment (DAVID), multi-omics analysis (RNA-seq, metabolomics, variant analysis), high-performance computing (TACC), containerized pipelines (BioContainers, Docker), version-controlled workflows (Git/GitHub)
- **Analytical Chemistry & Instrumentation**: HPLC, LC-MS/MS, GC-MS with FAME derivatization, UV-Vis, microscopy, rheology, COD and enzyme activity assays
- **Software & Reproducibility Tools**: Linux/Unix, Bash, Git/GitHub, Jupyter, Quarto, LaTeX, GraphPad Prism

Select Honors and Awards

- **First Place, Professional Division at the Data Challenge Expo, JSM (Joint Statistical Meetings)**, 2025
- **National Science Foundation Graduate Research Fellowship (NSF GRFP)**, 2021
- **Provost's Graduate Excellence Fellowship, UT Austin**, 2019
- **Individual Accomplishment Citation, MIT Department of Chemical Engineering**, 2018 & 2019
  Only undergraduate to win this award two years in a row, for volunteer work and service to the department

Publications

13 peer-reviewed publications and one preprint, seven as first or co-first author (* co-first). Selected:
- **Coleman, S. M.*** et al. High oil density bioreactor-scale fermentations of *Yarrowia lipolytica* using CFD modeling and experimental validation. *Biotechnology Journal* (2024)
- **Coleman, S. M.*** et al. Evolving tolerance of *Yarrowia lipolytica* to hydrothermal liquefaction aqueous phase waste. *Applied Microbiology and Biotechnology* (2023)

Full list: sarahmcoleman.github.io/papers.html